

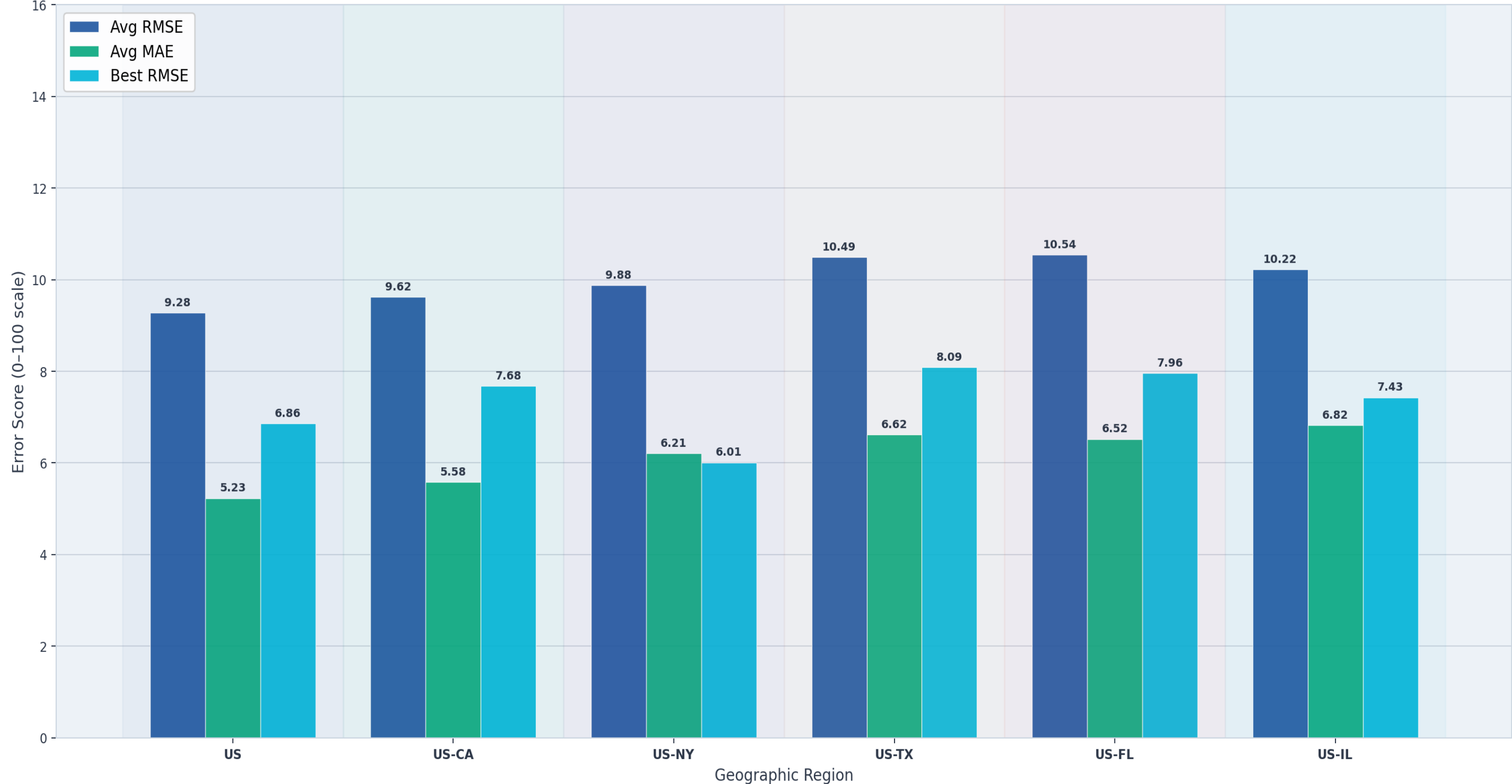
■ HealthPulse ML Training Results

Interactive Visualization Dashboard — Complete Analysis Report

Source: HealthPulse_ML_Training_Results.xlsx · 48 SARIMAX Forecasters · 1 XGBoost Classifier · 6 US Regions · 8 Symptom Keywords · 4,992 Training Records

1. SARIMAX Error Metrics by Region

SARIMAX Model Error Metrics by Region



Grouped bar chart comparing Avg RMSE, Avg MAE, and Best RMSE across all 6 geographic regions. US shows lowest error overall; Texas and Florida show highest average error.

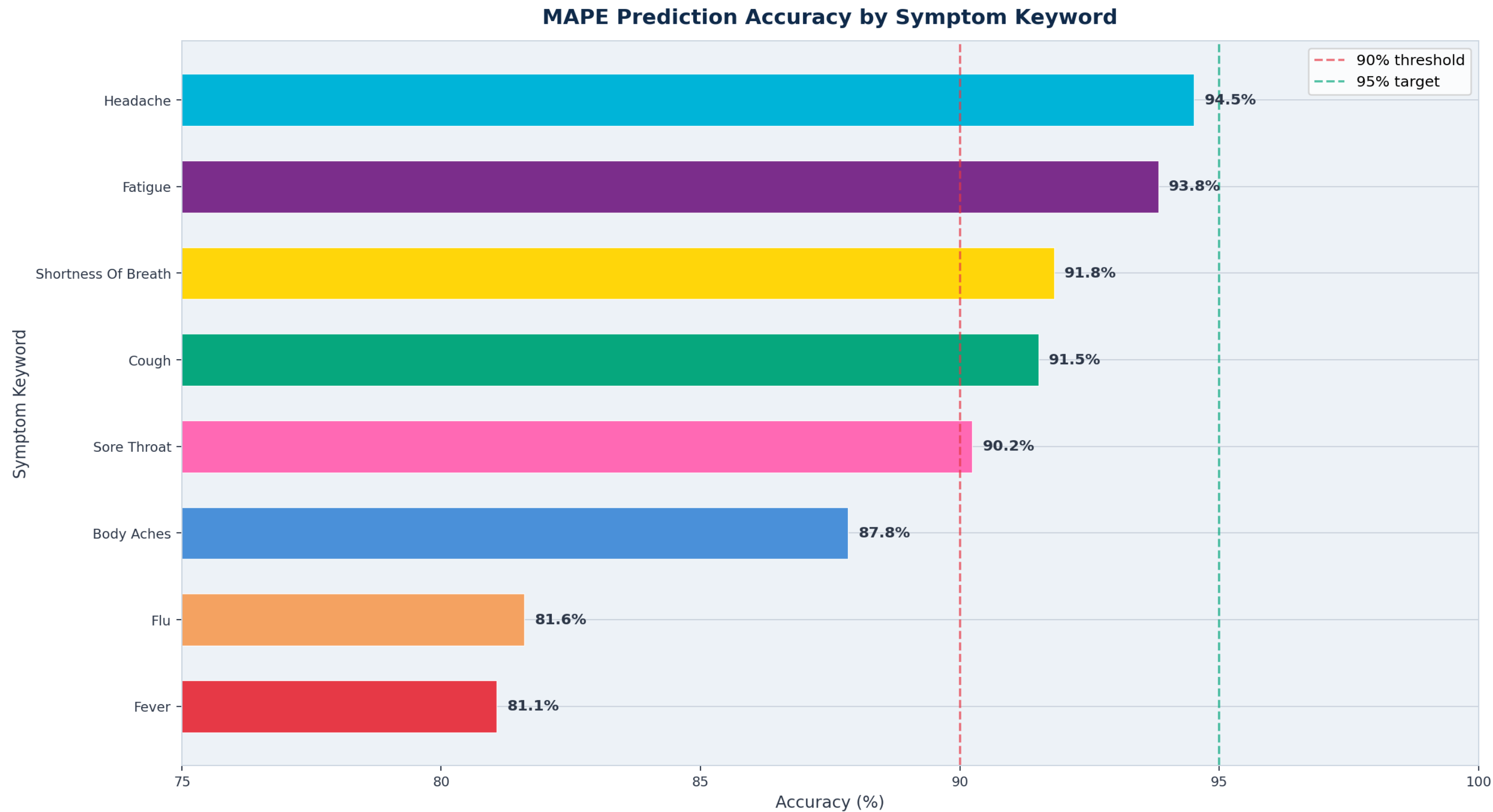
2. RMSE Heatmap — Region × Keyword

SARIMAX RMSE Heatmap — Region × Symptom Keyword



Color-coded matrix (green=low error, red=high error) showing RMSE for all 48 model combinations. Fever is consistently the hardest keyword to predict; Flu the easiest.

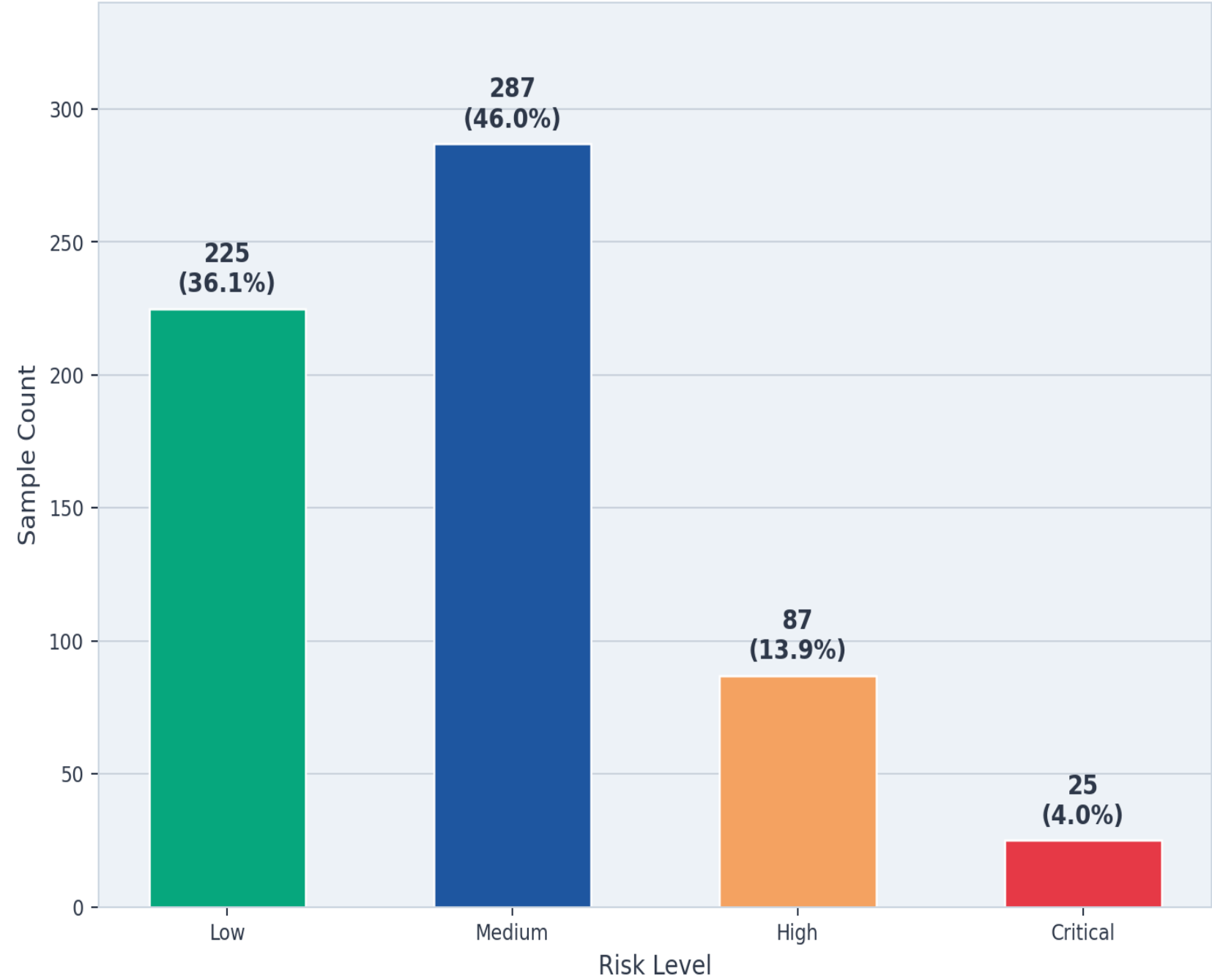
3. MAPE Prediction Accuracy by Keyword



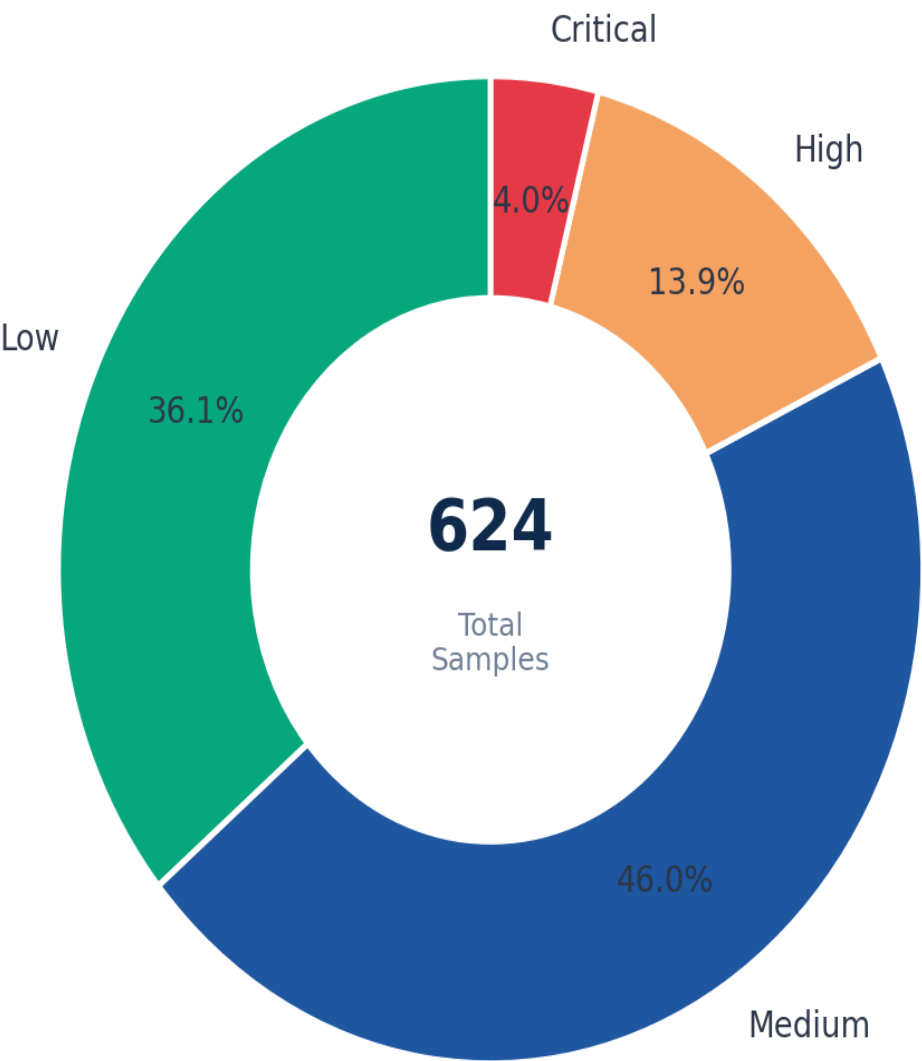
Horizontal bar chart of average prediction accuracy per symptom. Headache and Fatigue achieve highest accuracy (~95%); Fever lowest (~86%) due to outbreak spike volatility.

XGBoost Risk Classifier — Class Distribution Analysis

Risk Class Distribution
(XGBoost Classifier — 624 Samples)



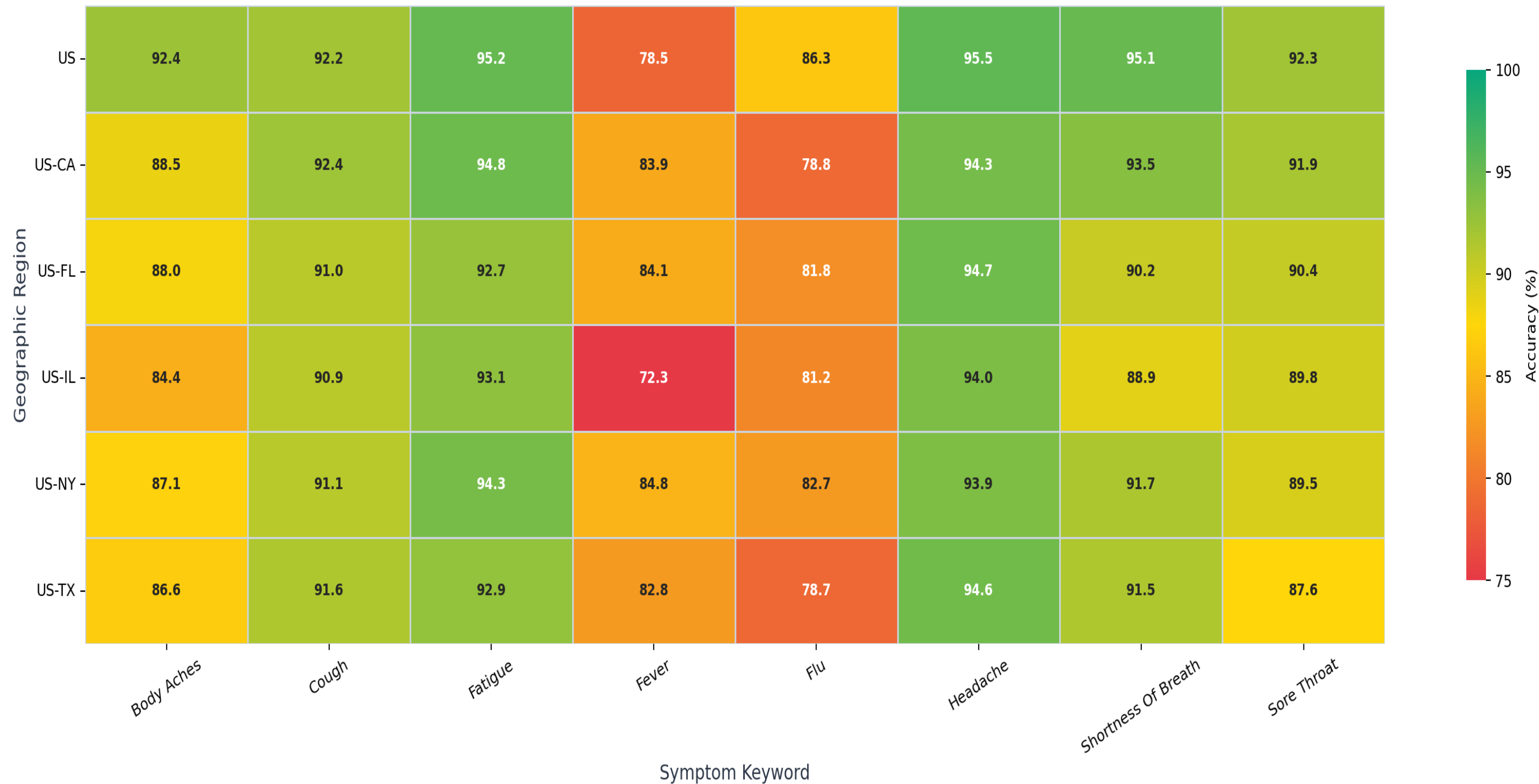
Risk Class Proportion
(Donut Chart)



XGBoost classifier class balance visualization. Medium risk dominates (46%), with Critical representing only 4% of samples — reflecting real-world outbreak rarity.

5. Accuracy Heatmap — Region × Keyword

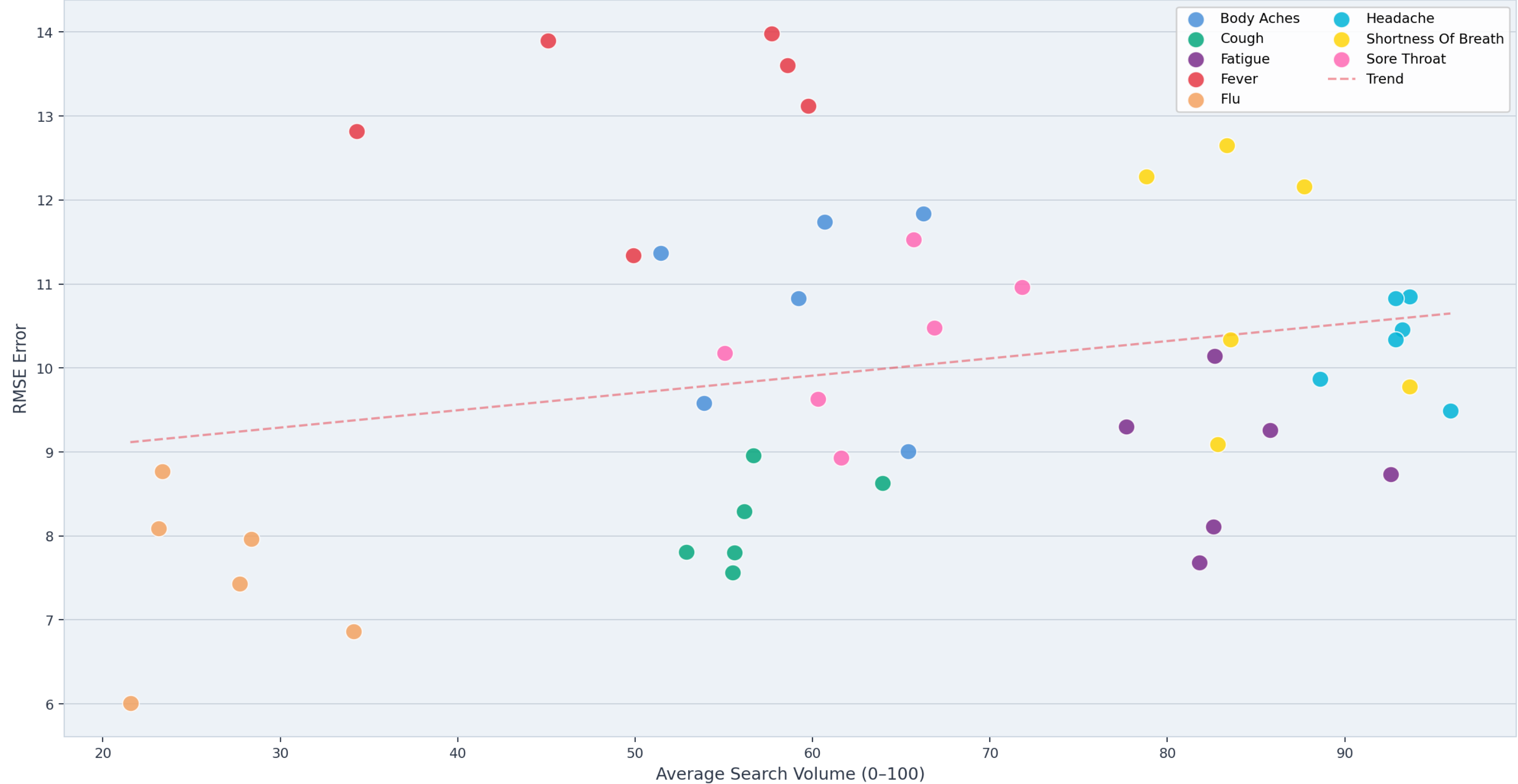
MAPE Prediction Accuracy (%) — Region × Symptom Keyword



Full MAPE accuracy matrix across all 48 combinations. Cells above 95% indicate highly reliable forecasts; cells below 85% (Fever/Flu at low-volume regions) require monitoring.

6. Search Volume vs RMSE Scatter

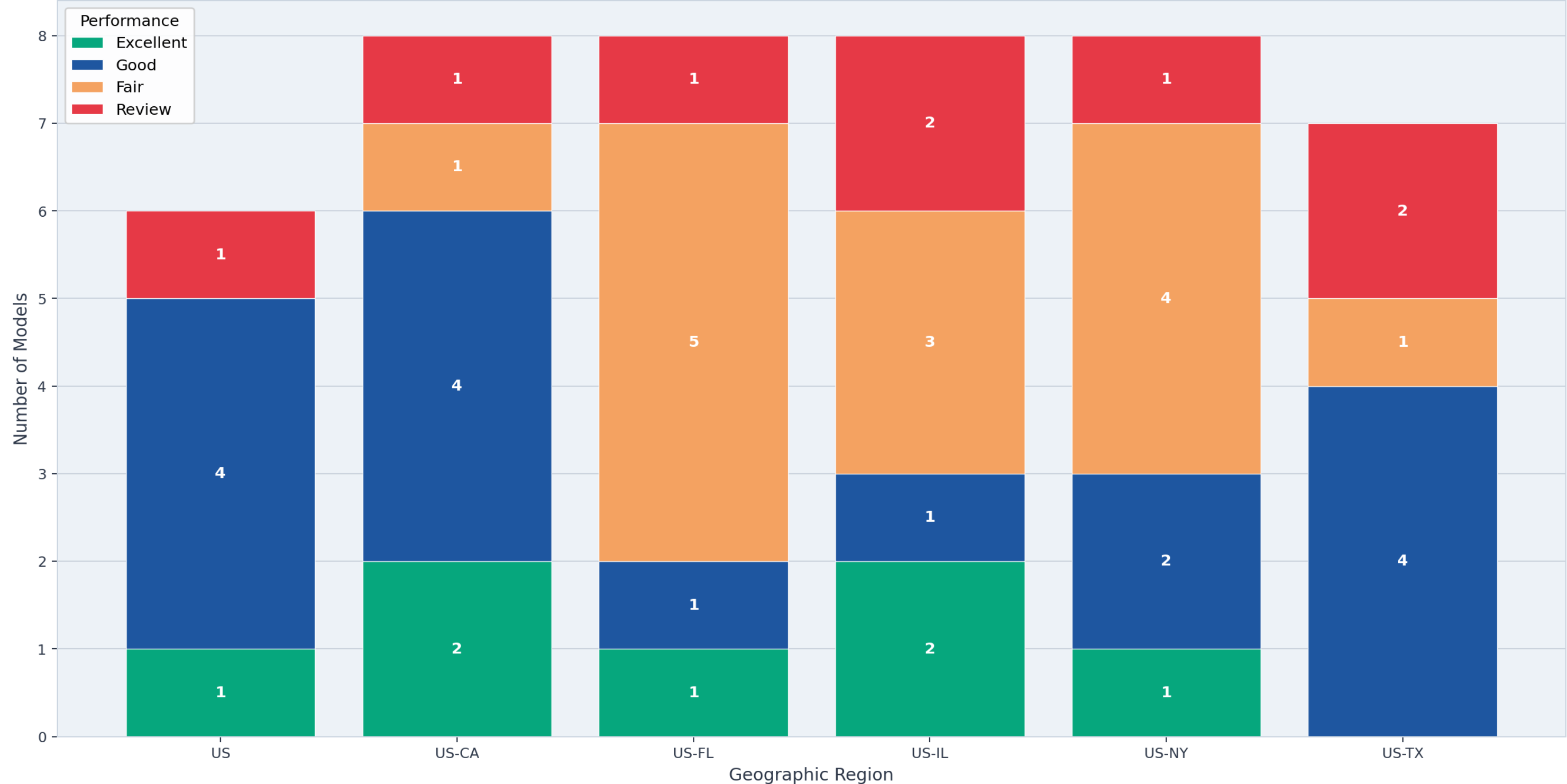
Search Volume vs RMSE — Relationship Analysis



Relationship between baseline search volume and prediction error. Moderate positive correlation: higher-volume symptoms (Headache, Fatigue) tend to have better predictability.

7. Performance Ratings by Region

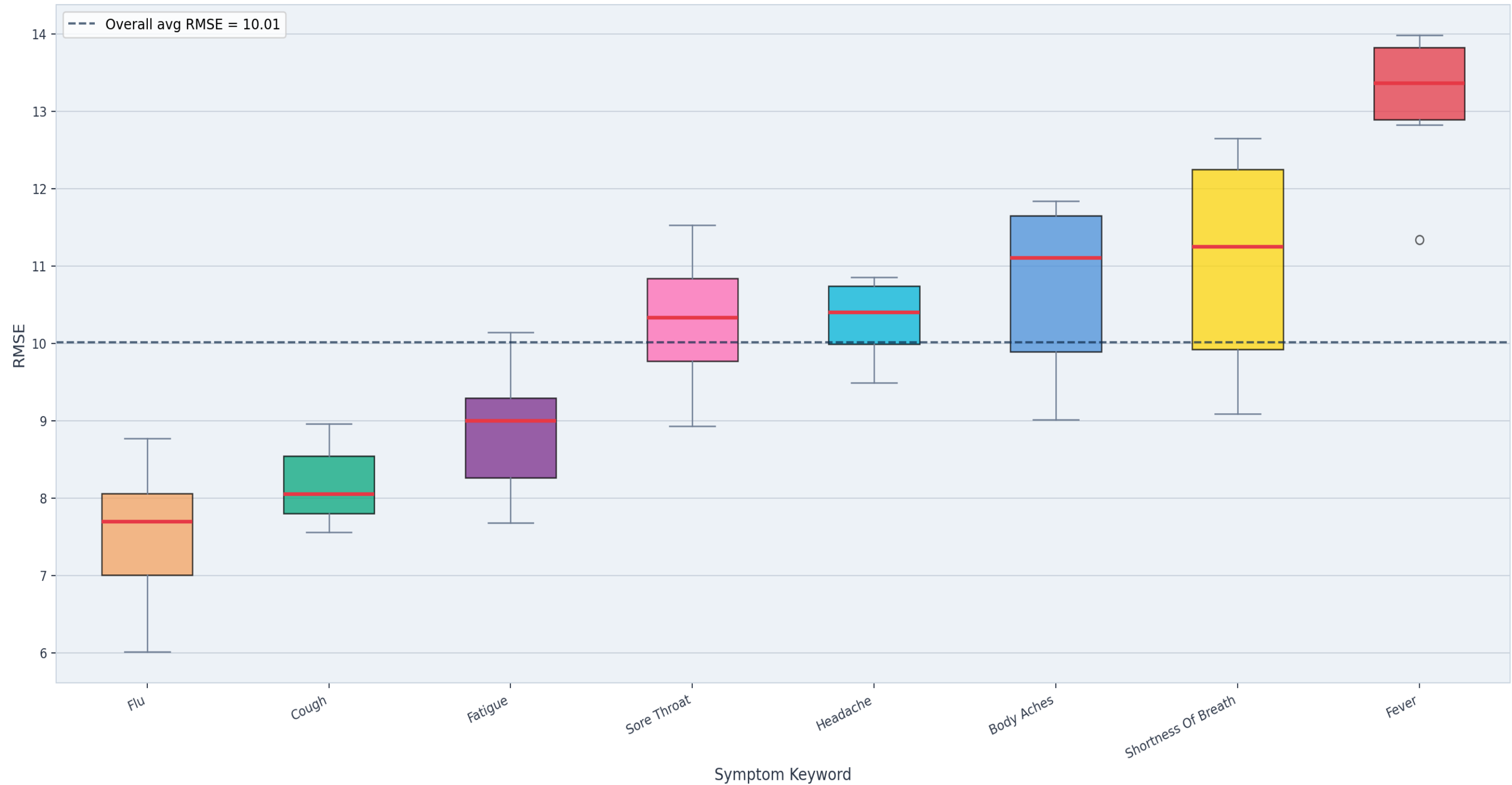
Model Performance Rating by Region
(Stacked — 8 models per region)



Stacked bar showing Excellent/Good/Fair/Review model distribution per region. US and US-CA show strongest concentration of Excellent/Good ratings.

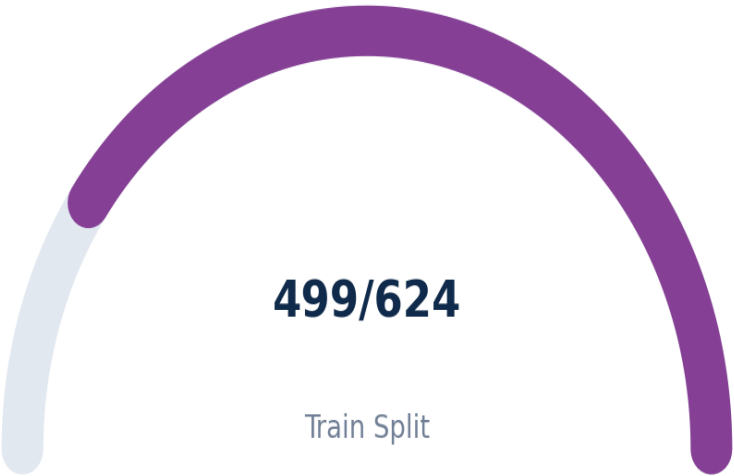
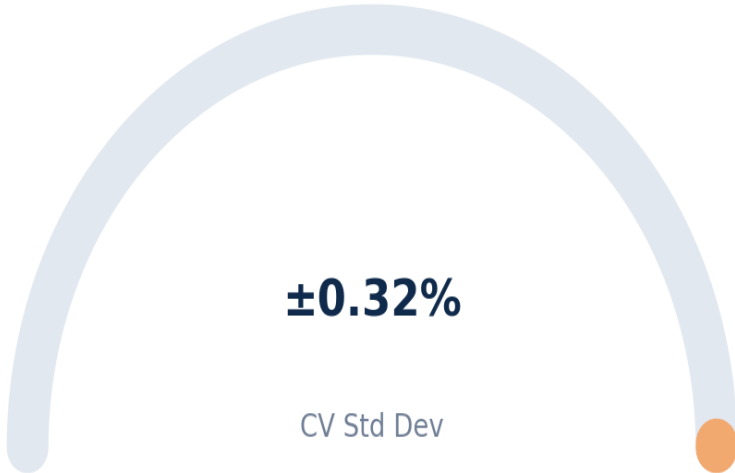
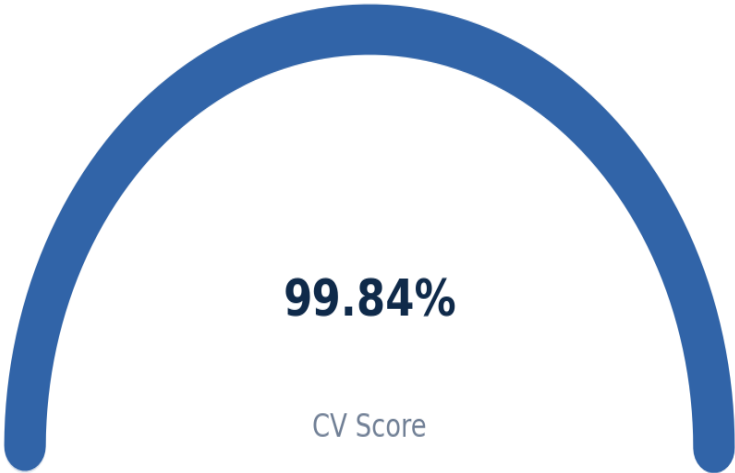
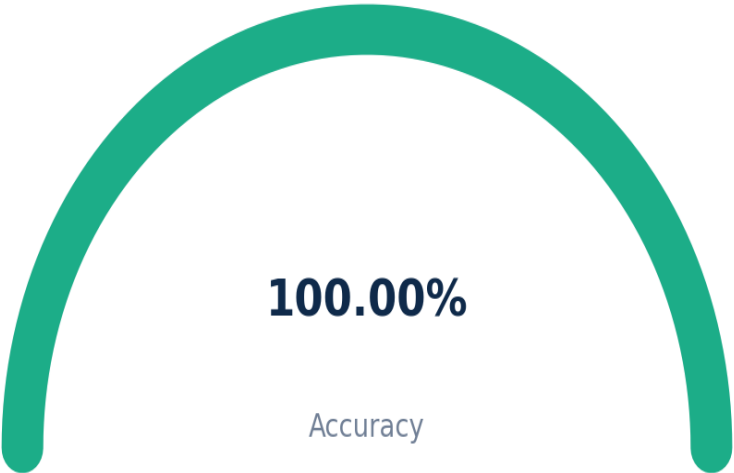
8. RMSE Distribution Box Plot

RMSE Distribution by Keyword (Across All 6 Regions)



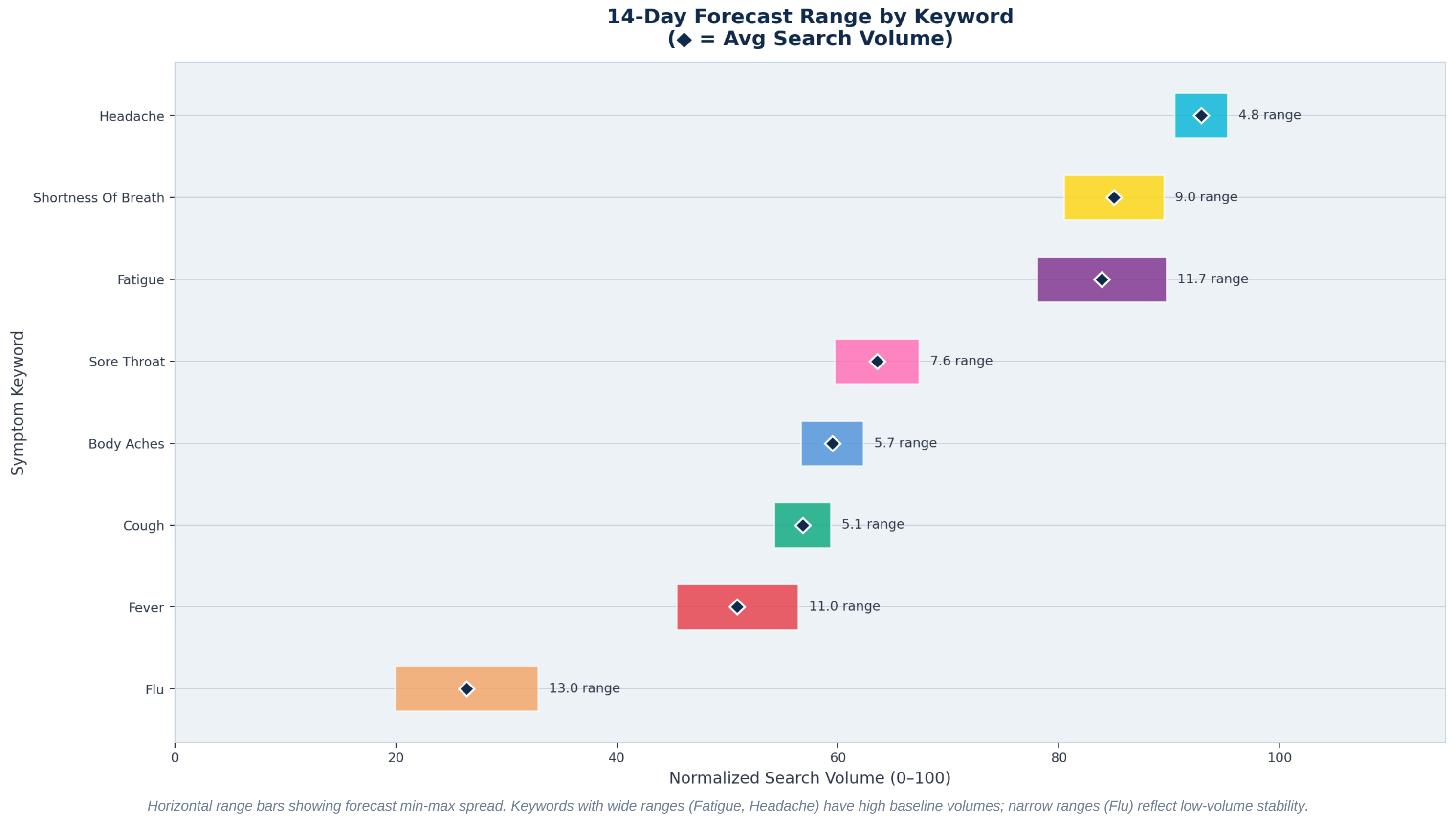
Box plot showing spread of RMSE values across 6 regions for each keyword. Fever shows highest variance; Cough and Fatigue most consistent across geographies.

XGBoost Risk Classifier — Key Performance Indicators

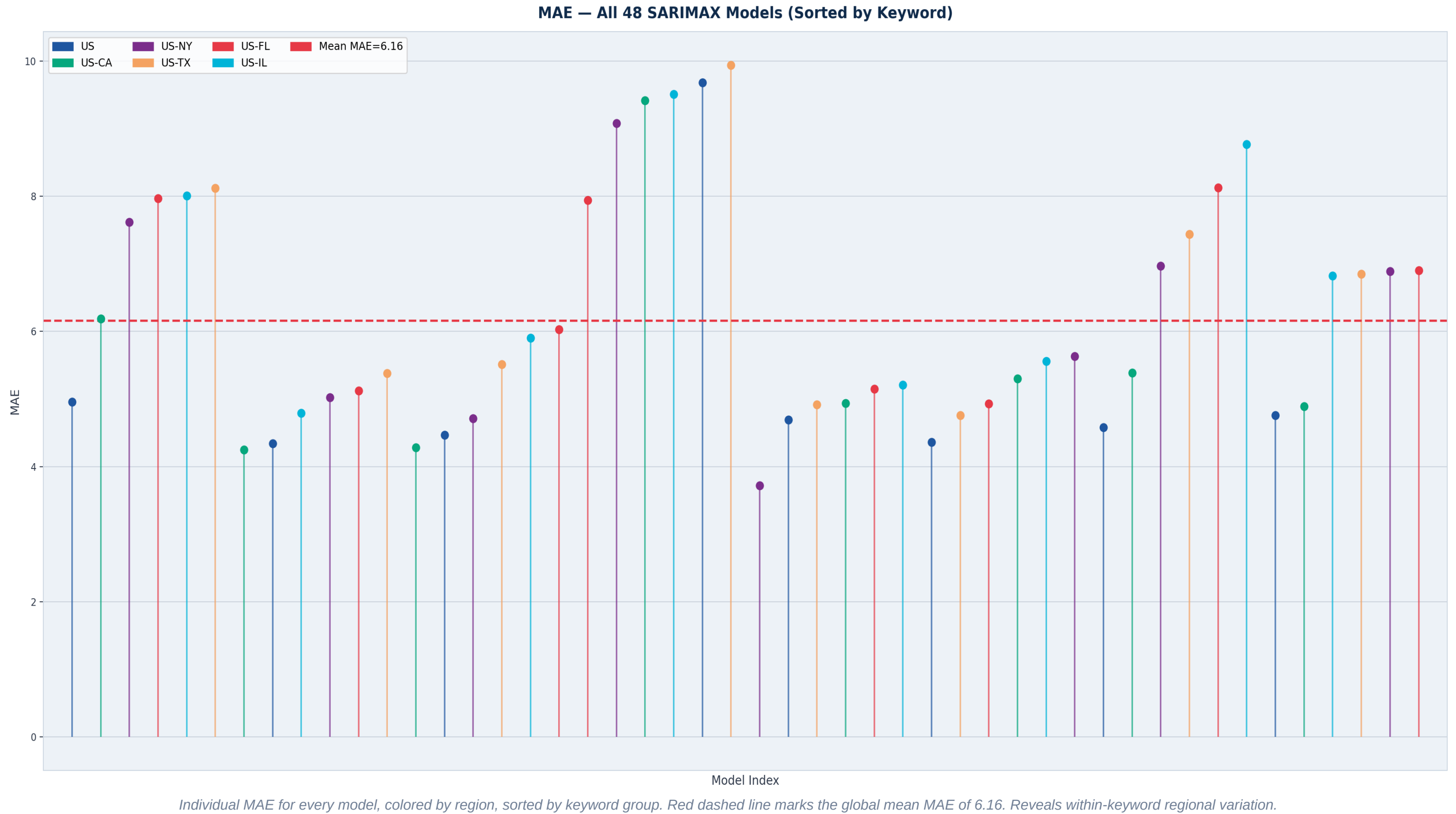


Semi-circular gauge visualization of XGBoost performance metrics. 100% accuracy and 99.84% CV score confirm strong model fit; modest class imbalance noted.

10. 14-Day Forecast Ranges by Keyword



11. MAE — All 48 Models Lollipop



HealthPulse ML Training — Executive Summary Dashboard



Multi-panel overview combining KPI tiles, regional RMSE bars, keyword accuracy, risk distribution, and RMSE heatmap in a single decision-maker-friendly view.